

Module 2

Transmission Line Theory

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Transmission Lines

A transmission line connects a generator to a load

OR, Transmission Lines Guide EM Waves within a system

Note: EM Waves traveling in a transmission line take the form of voltage/current waves

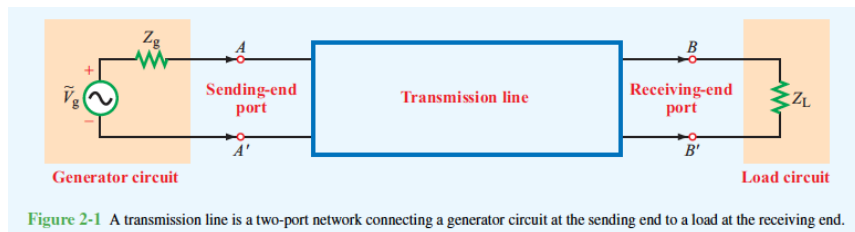


Figure 2-1 A transmission line is a two-port network connecting a generator circuit at the sending end to a load at the receiving end.

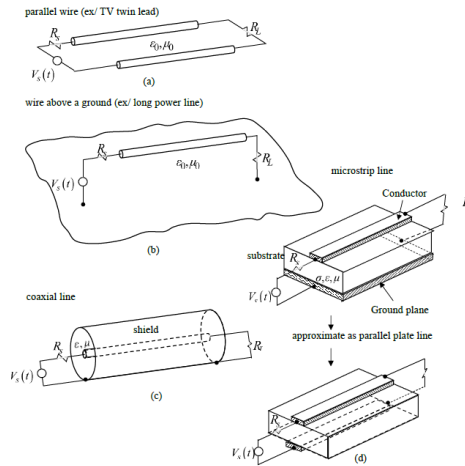
Transmission lines include:

- Two parallel wires
- Coaxial cable
- Microstrip line
- Optical fiber
- Waveguide

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Transmission Lines

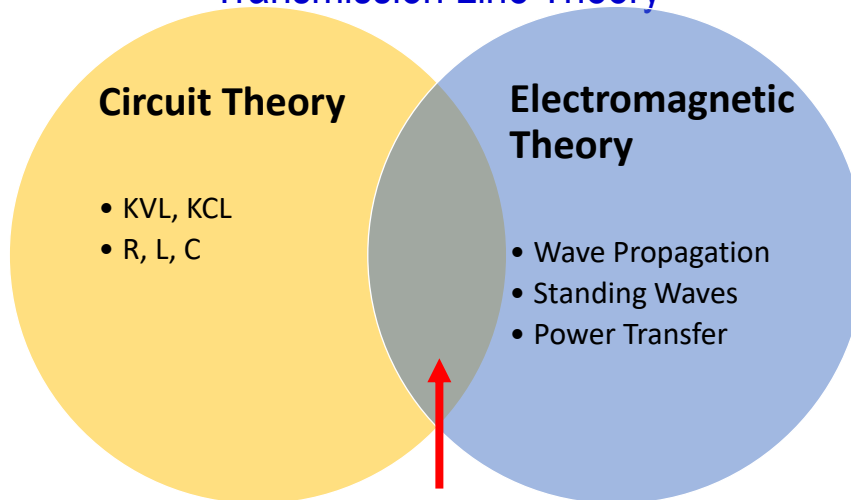
- This chapter focuses on transmission lines that guide electromagnetic signals.
- Such transmission lines include
 - telephone wires,
 - coaxial cables carrying audio and video information to TV sets or
 - digital data to computer monitors,
 - microstrips printed on microwave circuit boards,
 - and optical fibers carrying light waves for the transmission of data at **very high rates**.



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Transmission Line Theory



Transmission Line Theory
 “modeling transmission lines in the form of equivalent circuits”

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Here is what we will cover

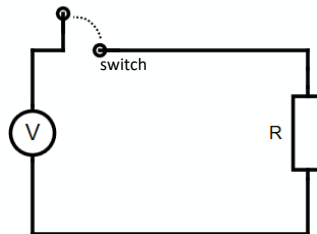
- Voltage/current wave propagation
- Characteristic Impedance
- Reflection and transmission
- Matching networks
- S-parameters

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Why talk about transmission lines?

- Electrically large circuits require more advanced analysis techniques than those taught for electrically-small systems.



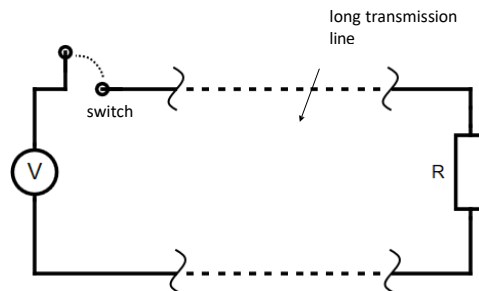
By conventional circuit theory (under the electrically-small-system assumption), closing the switch will instantly result in the voltage V appearing across the resistor R .

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Why talk about transmission lines?

- But suppose you have...



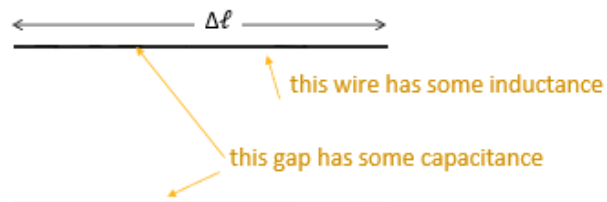
If the switch is far away from the load, then closing the switch will *not* instantaneously result in the voltage V appearing at the load. Rather, there will be a delay as the signal is transmitted down the line.

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The Telegrapher's Equations

- Let us look closely at a section of transmission line, of length $\Delta\ell$, where $\Delta\ell$ is electrically small ($\Delta\ell \ll \lambda$)

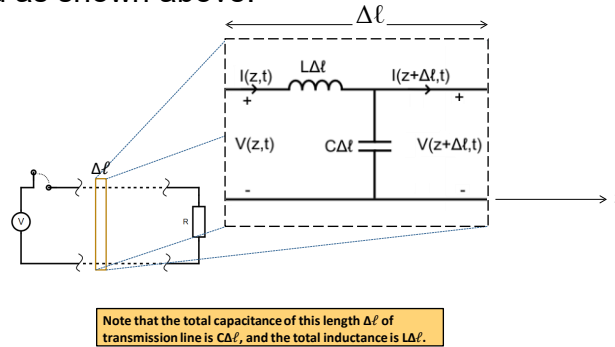


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The Telegrapher's Equations

- So, each piece of the long line has an inductance-per-length L and a capacitance-per-length C , which may be modeled as shown above.

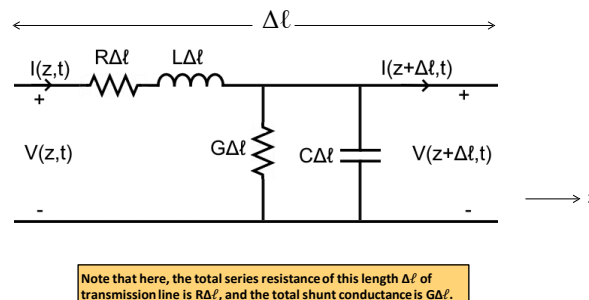


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The Telegrapher's Equations

- In the most general case, the line may also have resistive losses, which can be added to the model as:

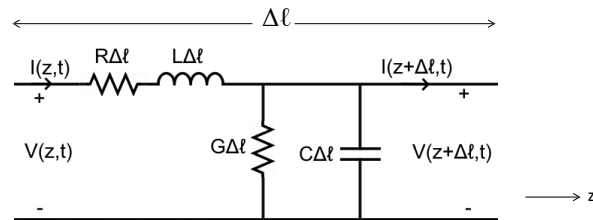


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The Telegrapher's Equations

- This system has input voltage $V(z,t)$, input current $I(z,t)$, output voltage $V(z+\Delta\ell,t)$ and output current $I(z+\Delta\ell,t)$.

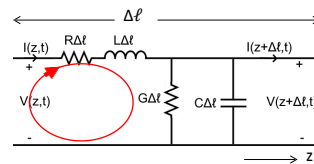


- Since we have assumed that $\Delta\ell$ is electrically small, we can analyze this circuit using normal circuit analysis.

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The Telegrapher's Equations



Using Kirchoff's voltage law, we can write:

$$V(z,t) - R\Delta\ell I(z,t) - L\Delta\ell \frac{dI(z,t)}{dt} - V(z+\Delta\ell,t) = 0$$

which can be rearranged to show:

$$\frac{V(z+\Delta\ell,t) - V(z,t)}{\Delta\ell} + L \frac{dI(z,t)}{dt} + RI(z,t) = 0$$

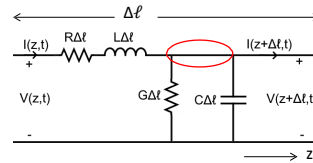
and, taking the limit as $\Delta\ell \rightarrow 0$, this becomes the spatial derivative:

$$\frac{dV(z,t)}{dz} + L \frac{dI(z,t)}{dt} + RI(z,t) = 0$$

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The Telegrapher's Equations



Similarly, by Kirchoff's current law:

$$I(z, t) - G\Delta\ell V(z + \Delta\ell, t) - C\Delta\ell \frac{dV(z + \Delta\ell, t)}{dt} - I(z + \Delta\ell, t) = 0$$

which can be rearranged to show:

$$\frac{I(z + \Delta\ell, t) - I(z, t)}{\Delta\ell} + C \frac{dV(z + \Delta\ell, t)}{dt} + GV(z + \Delta\ell, t) = 0$$

and, taking the limit as $\Delta\ell \rightarrow 0$, this becomes the spatial derivative:

$$\frac{dI(z, t)}{dz} + C \frac{dV(z, t)}{dt} + GV(z, t) = 0$$

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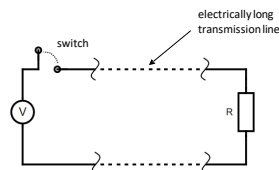
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The Telegrapher's Equations

$$\textcircled{1} \quad \frac{dV(z, t)}{dz} + L \frac{dI(z, t)}{dt} + RI(z, t) = 0$$

$$\textcircled{2} \quad \frac{dI(z, t)}{dz} + C \frac{dV(z, t)}{dt} + GV(z, t) = 0$$

These are the **telegrapher's equations**, which relate voltage and current along a transmission line.



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The Wave Equations

- The Telegrapher's Equations may be used to derive the wave equations for voltage and current along a transmission line.

A **wave equation** relates a quantity's second derivative in time to its second derivative in space.

Current and voltage on a transmission line may be described using wave equations, which can be derived from the telegrapher's equations as follows:

$$\textcircled{1} \quad \frac{dV(z,t)}{dz} = -L \frac{dI(z,t)}{dt} - RI(z,t) \quad \textcircled{2} \quad \frac{dI(z,t)}{dz} = -C \frac{dV(z,t)}{dt} - GV(z,t)$$

$$\frac{d}{dt} \textcircled{2} \longrightarrow \frac{d^2 I(z,t)}{dz dt} = -C \frac{d^2 V(z,t)}{dt^2} - G \frac{dV(z,t)}{dt}$$

$$\frac{d}{dz} \textcircled{1} \longrightarrow \frac{d^2 V(z,t)}{dz^2} = -L \frac{d^2 I(z,t)}{dz dt} - R \frac{dI(z,t)}{dz}$$

$$\frac{d^2 V(z,t)}{dz^2} = -L \left[-C \frac{d^2 V(z,t)}{dt^2} - G \frac{dV(z,t)}{dt} \right] - R \left[-C \frac{dV(z,t)}{dt} - GV(z,t) \right]$$

$$\frac{d^2 V(z,t)}{dz^2} = LC \frac{d^2 V(z,t)}{dt^2} + (LG + RC) \frac{dV(z,t)}{dt} + RGV(z,t) \longrightarrow \text{Wave equation for voltage} \quad 15$$

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The Wave Equations

- The Telegrapher's Equations may be used to derive the wave equations for voltage and current along a transmission line.

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$$\frac{d}{dt} \textcircled{1} \longrightarrow \frac{d^2 V(z,t)}{dz dt} = -L \frac{d^2 I(z,t)}{dt^2} - R \frac{dI(z,t)}{dt}$$

$$\frac{d}{dz} \textcircled{2} \longrightarrow \frac{d^2 I(z,t)}{dz^2} = -C \frac{d^2 V(z,t)}{dz dt} - G \frac{dV(z,t)}{dz}$$

$$\frac{d^2 I(z,t)}{dz^2} = -C \left[-L \frac{d^2 I(z,t)}{dt^2} - R \frac{dI(z,t)}{dt} \right] - G \left[-L \frac{dI(z,t)}{dt} - RI(z,t) \right]$$

$$\frac{d^2 I(z,t)}{dz^2} = LC \frac{d^2 I(z,t)}{dt^2} + (LG + RC) \frac{dI(z,t)}{dt} + RGI(z,t) \longrightarrow \text{Wave equation for current} \quad 16$$

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The Wave Equations

$$\frac{d^2V(z, t)}{dz^2} = LC \frac{d^2V(z, t)}{dt^2} + (LG + RC) \frac{dV(z, t)}{dt} + RGV(z, t)$$

This is the wave equation for **voltage**, which relates its second derivative in **time** to its second derivative in **space**.

$$\frac{d^2I(z, t)}{dz^2} = LC \frac{d^2I(z, t)}{dt^2} + (LG + RC) \frac{dI(z, t)}{dt} + RGI(z, t)$$

This is the wave equation for **current**, which relates its second derivative in **time** to its second derivative in **space**.

Note that the wave equations for voltage and current are **identical**. These differential equations will also have identical solutions.

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Special Case – Lossless Transmission Lines

- For a lossless line, the telegrapher's equations and wave equations can be simplified.

On a lossless transmission line, the series resistance vanishes ($R=0$), and the shunt conductance vanishes ($G=0$). For this case:

$$\frac{dV(z, t)}{dz} = -L \frac{dI(z, t)}{dt} \quad \frac{dI(z, t)}{dz} = -C \frac{dV(z, t)}{dt}$$

The Lossless Telegrapher's Equations

$$\frac{d^2V(z, t)}{dz^2} = LC \frac{d^2V(z, t)}{dt^2} \quad \frac{d^2I(z, t)}{dz^2} = LC \frac{d^2I(z, t)}{dt^2}$$

The Lossless Wave Equations

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Wave Equation Solutions

- Current and voltage will propagate on the line according to the solutions to the wave equations.

We will assume that both voltage and current are time-harmonic ($\sim e^{j\omega t}$). In other words, we assume that:

$$V(z, t) = V(z)e^{j\omega t}$$

$$I(z, t) = I(z)e^{j\omega t}$$

This allows us to write the wave equations as:

$$\frac{d^2 V(z)}{dz^2} e^{j\omega t} = -\omega^2 L C V(z) e^{j\omega t} + j\omega (L G + R C) V(z) e^{j\omega t} + R G V(z) e^{j\omega t}$$

$$\frac{d^2 I(z)}{dz^2} e^{j\omega t} = -\omega^2 L C I(z) e^{j\omega t} + j\omega (L G + R C) I(z) e^{j\omega t} + R G I(z) e^{j\omega t}$$

Or, by rearranging:

$$\frac{d^2 V(z)}{dz^2} = V(z) [R G + j\omega (L G + R C) - \omega^2 L C]$$

$$\frac{d^2 I(z)}{dz^2} = I(z) [R G + j\omega (L G + R C) - \omega^2 L C]$$

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Propagation Constant

- We will pause here to define the complex propagation constant

The complex propagation constant γ is defined as:

$$\gamma^2 = R G + j\omega (L G + R C) - \omega^2 L C \quad \text{or} \quad \gamma = \sqrt{R G + j\omega (L G + R C) - \omega^2 L C}$$

We will also define the **attenuation constant** α and the **lossless propagation constant** β as the real and imaginary parts of γ , respectively.

$$\gamma = \alpha + j\beta$$

Note that, in the lossless case,

$$\gamma = j\omega\sqrt{LC}$$

$$\gamma = \alpha + j\beta$$

$$\gamma = j\beta$$

where

$$\alpha = 0$$

$$\text{and } \beta = \omega\sqrt{LC}$$

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Wave Equation Solutions

- Current and voltage will propagate on the line according to the solutions to the wave equations.

Using our definition of the complex propagation constant, we can rewrite the wave equations as:

$$\frac{d^2 V(z)}{dz^2} = \gamma^2 V(z)$$

$$\frac{d^2 I(z)}{dz^2} = \gamma^2 I(z)$$

which are differential equations with solutions of either sines and cosines or complex exponentials. We will choose to use complex exponentials.

$$\begin{aligned} V(z) &= V_o^+ e^{-\gamma z} + V_o^- e^{+\gamma z} \\ I(z) &= I_o^+ e^{-\gamma z} + I_o^- e^{+\gamma z} \end{aligned}$$

where V_o^+ , V_o^- , I_o^+ , and I_o^- are variables corresponding to magnitude.

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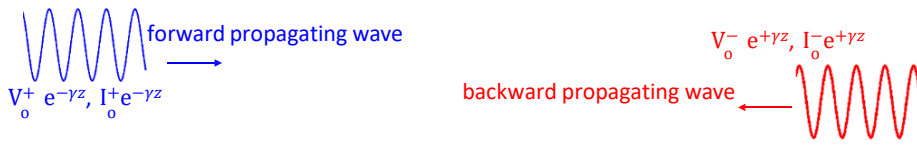
Wave Equation Solutions

$$V(z) = V_o^+ e^{-\gamma z} + V_o^- e^{+\gamma z}$$

$$I(z) = I_o^+ e^{-\gamma z} + I_o^- e^{+\gamma z}$$

Each of these solutions consists of two terms. The **terms including $e^{-\gamma z}$** propagate in the **+z direction** (forward propagation), and the **terms including $e^{+\gamma z}$** propagate in the **-z direction** (backward propagation).

→ Z



The total voltage on the transmission line is the sum of the forward-propagating part and the backward propagating part.

The total current is similarly the sum of the forward and backward propagating terms.

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Wave Equation Solutions

Recall: we said earlier that

$$V(z, t) = V(z)e^{j\omega t}$$

$$I(z, t) = I(z)e^{j\omega t}$$

so our total solutions for voltage and current are:

$$V(z, t) = V_0^+ e^{-\gamma z} e^{j\omega t} + V_0^- e^{+\gamma z} e^{j\omega t}$$

$$I(z, t) = I_0^+ e^{-\gamma z} e^{j\omega t} + I_0^- e^{+\gamma z} e^{j\omega t}$$

but this is commonly written in phasor form, which suppresses the harmonic term:

$$\begin{aligned} V(z, t) &= V_0^+ e^{-\gamma z} + V_0^- e^{+\gamma z} = V_0^+ e^{-\alpha z} e^{-j\beta z} + V_0^- e^{+\alpha z} e^{+j\beta z} \\ I(z, t) &= I_0^+ e^{-\gamma z} + I_0^- e^{+\gamma z} = I_0^+ e^{-\alpha z} e^{-j\beta z} + I_0^- e^{+\alpha z} e^{+j\beta z} \end{aligned}$$

or it may be written equivalently in the time domain:

$$\begin{aligned} V(z, t) &= V_0^+ e^{-\alpha z} \cos(\omega t - \beta z) + V_0^- e^{+\alpha z} \cos(\omega t + \beta z) \\ I(z, t) &= I_0^+ e^{-\alpha z} \cos(\omega t - \beta z) + I_0^- e^{+\alpha z} \cos(\omega t + \beta z) \end{aligned}$$

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Characteristic Impedance

- *The ratio of voltage to current at any point along a transmission line is fixed by the characteristics of the line.*

Let us look at just the forward-propagating components of voltage and current, in phasor form:

$$V(z, t) = V_0^+ e^{-\gamma z}$$

$$I(z, t) = I_0^+ e^{-\gamma z}$$

and recall the first of the telegrapher's equations:

$$\frac{dV(z, t)}{dz} = -L \frac{dI(z, t)}{dt} - RI(z, t)$$

We can combine these equations as follows:

$$\frac{d}{dz} [V_0^+ e^{-\gamma z}] = -L \frac{d}{dt} [I_0^+ e^{-\gamma z}] - R [I_0^+ e^{-\gamma z}]$$

$$-\gamma V_0^+ e^{-\gamma z} = -j\omega L I_0^+ e^{-\gamma z} - R I_0^+ e^{-\gamma z}$$

which can be rearranged to give the characteristic impedance, defined by:

$$Z_o = \frac{V_0^+}{I_0^+} = \frac{j\omega L + R}{\gamma} = \frac{j\omega L + R}{\sqrt{RG + j\omega(LG + RC) - \omega^2 LC}} = \sqrt{\frac{R + j\omega L}{G + j\omega C}}$$

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Characteristic Impedance

This is the characteristic impedance of the line, given in terms of its per-length resistance, inductance, conductance, and capacitance.

$$Z_o = \frac{V_o^+}{I_o^+} = \sqrt{\frac{R + j\omega L}{G + j\omega C}}$$

Note that, if the line is lossless, this becomes:

$$Z_o = \sqrt{L/C}$$

Note that the characteristic impedance of a lossless line is purely real.

The characteristic impedance is the ratio of forward voltage to forward current. The same derivation may be performed in terms of the backward voltage and backward current, to show:

$$Z_o = \sqrt{\frac{R + j\omega L}{G + j\omega C}} = \frac{\text{forward voltage}}{\text{forward current}} = - \frac{\text{backward voltage}}{\text{backward current}}$$

Note the negative sign of the backward terms

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Summary

Angular Frequency: $\omega = 2\pi f$

Complex Propagation Constant: $\gamma = \sqrt{RG + j\omega(LG + RC) - \omega^2 LC} = \alpha + j\beta$

Attenuation Constant: $\alpha = \text{Re}\{\gamma\}$

Lossless Propagation Constant: $\beta = \text{Im}\{\gamma\} = \frac{2\pi}{\lambda}$

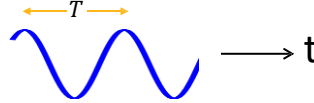
Characteristic Impedance: $Z_o = \sqrt{\frac{R + j\omega L}{G + j\omega C}}$

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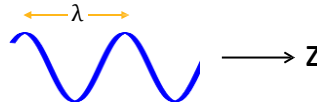
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Summary

Period: $T = 1/f$ = time in seconds between two peaks of a wave at a given location in space



Wavelength: λ = distance in meters between two peaks of a wave at a given instant in time



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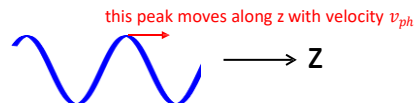
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Wave Properties – Phase Velocity

Phase velocity:

$$v_{ph} = \frac{\omega}{\beta}$$

= velocity of the **phase front** of the wave



Note that in a lossy medium, the phase velocity is a function of frequency. This leads to **dispersion**, where a wave packet containing a range of frequencies (such as a square wave) input at one end will lose its shape over the length of the line, as the various frequencies travel at different rates.

In a **lossless** medium ($R=G=0$), the frequency-dependence disappears, and we are left with:

$$v_{ph} = \frac{1}{\sqrt{LC}}$$

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Transmission Line Effects - Dispersion

A dispersive transmission line is one on which the wave velocity is not constant as a function of the frequency

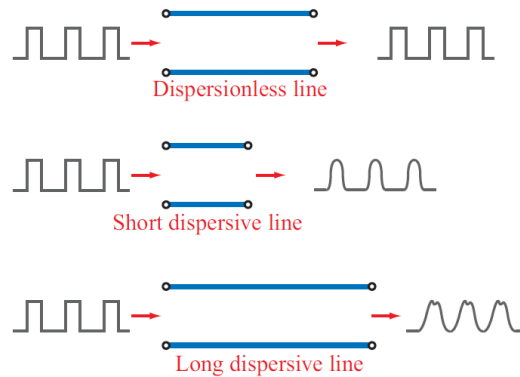
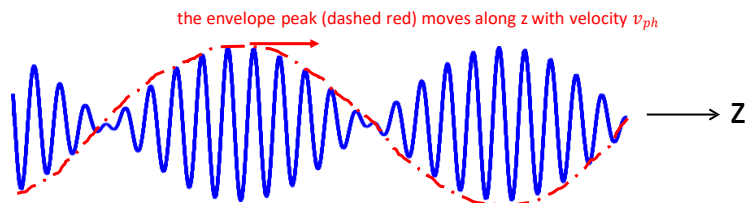


Figure 2-3: A dispersionless line does not distort signals passing through it regardless of its length, whereas a dispersive line distorts the shape of the input pulses because the different frequency components propagate at different velocities. The degree of distortion is proportional to the length of the dispersive line.

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Wave Properties – Group Velocity

Group velocity: $v_g = \frac{\delta\omega}{\delta\beta}$ (this definition holds for **narrowband** signals)
= velocity of the **envelope** of the wave



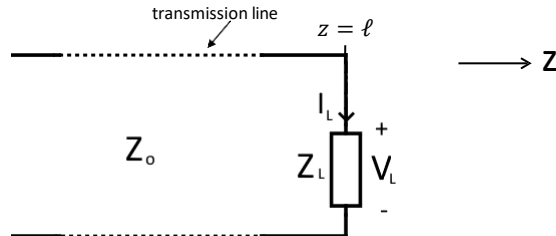
Note that if the wave consists of a single frequency, the group and phase velocities are equal.

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Transmission Line with a Load

- Let us look at what happens when a lossless transmission line is used to drive a load at location $z = \ell$



At the load end, the ratio of total voltage to total current must conform to Ohm's Law

$$Z_L = \frac{V_L}{I_L} = \frac{V(\ell)}{I(\ell)} = \frac{V_o^+ e^{-j\beta\ell} + V_o^- e^{+j\beta\ell}}{I_o^+ e^{-j\beta\ell} + I_o^- e^{+j\beta\ell}}$$

On the line, the total voltage and total current consist of forward traveling and backward traveling components, which are related by:

$$Z_o = \frac{V_o^+}{I_o^+} = -\frac{V_o^-}{I_o^-}$$

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Transmission Line with a Load

$$Z_L = \frac{V_L}{I_L} = \frac{V(\ell)}{I(\ell)} = \frac{V_o^+ e^{-j\beta\ell} + V_o^- e^{-j\beta\ell}}{I_o^+ e^{-j\beta\ell} + I_o^- e^{-j\beta\ell}}$$

$$Z_o = \frac{V_o^+}{I_o^+} = -\frac{V_o^-}{I_o^-}$$

These two relationships may be rearranged to show that:

$$\frac{V_o^-}{V_o^+} = \frac{Z_L e^{-j\beta\ell} - Z_o e^{-j\beta\ell}}{Z_L e^{+j\beta\ell} + Z_o e^{+j\beta\ell}}$$

And we can arbitrarily assign $z = \ell = 0$ as the zero location for the z -axis. This leads to the following definition of the reflection coefficient:

$$\Gamma = \frac{V_o^-}{V_o^+} = \frac{Z_L - Z_o}{Z_L + Z_o}$$

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Reflection Coefficient

$$\Gamma = \frac{V_o^-}{V_o^+} = \frac{Z_L - Z_o}{Z_L + Z_o}$$

If a transmission line is driven with input voltage V_o^+ , the signal will travel down the line to the load. Once it reaches the load, there are two possible routes it may follow. The signal may flow into the load, or it may reflect back toward the source. The reflection coefficient gives the ratio of the reflected voltage to the input voltage.

Consider three special cases.

- 1) $Z_L = Z_o$ The is the perfect match scenario. In this case, $\Gamma = 0$. All the signal flows into the load.
- 2) $Z_L \rightarrow \infty$ The is the open circuit scenario. In this case, $\Gamma = 1$. All the signal reflects back toward the source.
- 3) $Z_L = 0$ The is the short circuit scenario. In this case, $\Gamma = -1$. All the signal reflects back toward the source, with a phase shift of 180° .

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Reflection Coefficient

$$\Gamma = \frac{V_o^-}{V_o^+} = \frac{Z_L - Z_o}{Z_L + Z_o}$$

Note: On a lossless line, the characteristic impedance is positive and purely real. If the load is passive ($\text{Re}\{Z_L\} \geq 0$), then the magnitude of the reflection coefficient is between zero and one ($0 \leq |\Gamma| \leq 1$)

We will define the standing wave ratio (SWR) on the line as:

$$\text{SWR} = \frac{1 + |\Gamma|}{1 - |\Gamma|}$$

and the return loss (RL) as:

$$\text{RL} = -20 \log_{10} |\Gamma|, \text{ dB}$$

Note that Γ , SWR, and RL are all various expressions of reflection.

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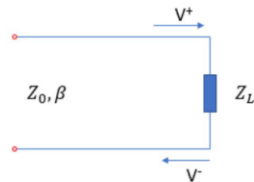
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Power

$$\square \frac{Z_L}{Z_0} = \overline{Z}_L = \frac{1 + \frac{V^-}{V^+}}{(1 - \frac{V^-}{V^+})}$$

$$\overline{Z}_L = \frac{1 + \Gamma}{1 - \Gamma} \quad *$$

$$\Gamma = \frac{V^-}{V^+}, \text{ the reflection Coefficient}$$



$$P_L = \frac{1}{2} \operatorname{Re}(V_L \cdot I_L^*)$$

$$= \frac{1}{2} \operatorname{Re}[(V^+ + V^-) \cdot (\frac{V^+}{Z_0} - \frac{V^-}{Z_0})^*]$$

$$= \frac{1}{2Z_0} \operatorname{Re}[V^+(1 + \Gamma) \cdot V^{+*}(1 - \Gamma)^*]$$

$$\text{Set } \Gamma = \rho e^{j\theta}, \quad |\Gamma| = \rho$$

$$P_L = \frac{1}{2} \frac{|V^+|^2}{Z_0} [1 - \rho^2]$$

$$P_L = P_{in}[1 - \rho^2] \quad *$$

↑ Power delivered to the load
↑ Incident power to the load

The standard impedance is 50 ohm.

When $Z_L = Z_0 \Rightarrow \Gamma = 0$, matched loads

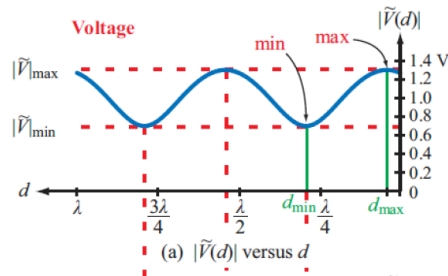
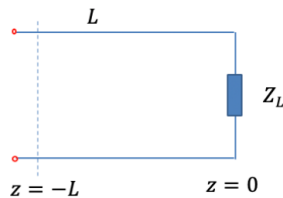
When $|\Gamma| = \rho = 0 \Rightarrow P_L = P_{in}$

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Standing Wave Ratio

Voltage and Transmission Lines



$$V = V^+ e^{-j\beta z} + V^- e^{j\beta z}$$

$$V = V^+[e^{-j\beta z} + \rho e^{j\theta} e^{j\beta z}]$$

$$|V|_{z=-L} = |V^+| \cdot |1 + \rho e^{j(\theta - 2\beta L)}|$$

$$|V|_{z=-L} = |V^+| \cdot [(1 + \rho)^2 - 4\rho \sin^2(\beta L - \frac{\theta}{2})]^{1/2}$$

$$V \text{ is max when } \beta L - \frac{\theta}{2} = n\pi$$

$$V_{max} = |V^+|(1 + \rho)$$

$$V \text{ is min when } \beta L - \frac{\theta}{2} = (n + \frac{1}{2})\pi$$

$$V_{min} = |V^+|(1 - \rho)$$

$$SWR = \frac{|V_{max}|}{|V_{min}|} = \frac{1 + \rho}{1 - \rho}$$

$$\rho = |\Gamma|$$

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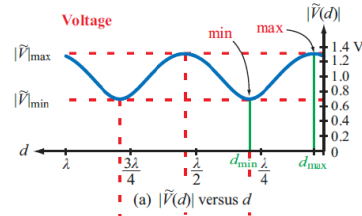
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Standing Wave Ratio

$$SWR = \frac{|V_{max}|}{|V_{min}|} = \frac{1 + \rho}{1 - \rho}$$

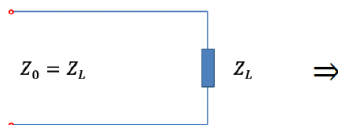
$$\rho = |\Gamma|$$

$$\Gamma = \rho e^{j\theta} = \frac{Z_L - Z_0}{Z_L + Z_0}$$



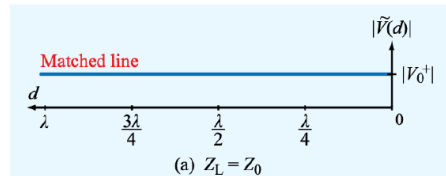
VSWR: Voltage Standing Wave Ratio, sometimes referred as SWR

$$Z_L = Z_0, \quad \Gamma = 0, \quad SWR = 1:$$



For a matched load: **SWR = 1**

- $SWR \gg 1$
- Devices with SWR of 1.05 or 1.1 are considered good. High SWR means the device is poorly matched



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Standing Wave Ratio

$$|V| = |V^+| \cdot [(1 + \rho)^2 - 4\rho \sin^2(\beta L - \frac{\theta}{2})]^{1/2}$$

$$\text{Short Circuit, } Z_L = 0, \quad \Gamma = -1$$

$$\text{Open Circuit } Z_L = \infty, \quad \Gamma = 1$$

low-frequency, if $f = 60 \text{ Hz}$,

$$\lambda = \frac{c}{f} = \frac{3 \cdot 10^8 \text{ m/s}}{60 \text{ Hz}} =$$

$$5 \cdot 10^6 \text{ m} = 5000 \text{ km}$$

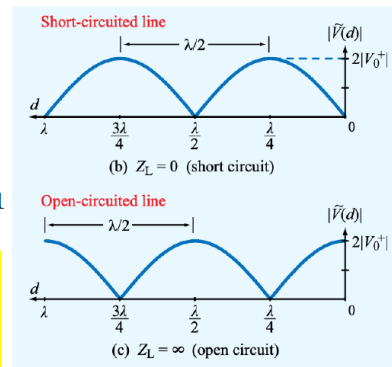
$$\frac{\lambda}{4} = 1250 \text{ km}$$

RF, if $f = 10 \text{ GHz}$,

$$\lambda = \frac{c}{f} = \frac{3 \cdot 10^8 \text{ m/s}}{10 \cdot 10^9 \text{ Hz}} =$$

$$= 30 \text{ mm}$$

$$\frac{\lambda}{4} = 7.5 \text{ mm}$$



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Standing Wave Summary

Table 2-4: Properties of standing waves on a lossless transmission line.

Voltage Maximum	$ \tilde{V} _{\max} = V_0^+ [1 + \Gamma]$
Voltage Minimum	$ \tilde{V} _{\min} = V_0^+ [1 - \Gamma]$
Positions of voltage maxima (also positions of current minima)	$d_{\max} = \frac{\theta_t \lambda}{4\pi} + \frac{n\lambda}{2}, \quad n = 0, 1, 2, \dots$
Position of first maximum (also position of first current minimum)	$d_{\max} = \begin{cases} \frac{\theta_t \lambda}{4\pi}, & \text{if } 0 \leq \theta_t \leq \pi \\ \frac{\theta_t \lambda}{4\pi} + \frac{\lambda}{2}, & \text{if } -\pi \leq \theta_t \leq 0 \end{cases}$
Positions of voltage minima (also positions of current maxima)	$d_{\min} = \frac{\theta_t \lambda}{4\pi} + \frac{(2n+1)\lambda}{4}, \quad n = 0, 1, 2, \dots$
Position of first minimum (also position of first current maximum)	$d_{\min} = \frac{\lambda}{4} \left(1 + \frac{\theta_t}{\pi} \right)$
Input Impedance	$Z_{\text{in}} = Z_0 \left(\frac{Z_L + j \tan \beta l}{1 + j Z_L \tan \beta l} \right) = Z_0 \left(\frac{1 + \Gamma_l}{1 - \Gamma_l} \right)$
Positions at which Z_{in} is real	at voltage maxima and minima
Z_{in} at voltage maxima	$Z_{\text{in}} = Z_0 \left(\frac{1 + \Gamma }{1 - \Gamma } \right)$
Z_{in} at voltage minima	$Z_{\text{in}} = Z_0 \left(\frac{1 - \Gamma }{1 + \Gamma } \right)$
Z_{in} of short-circuited line	$Z_{\text{in}}^{\text{sc}} = j Z_0 \tan \beta l$
Z_{in} of open-circuited line	$Z_{\text{in}}^{\text{oc}} = -j Z_0 \cot \beta l$
Z_{in} of line of length $l = n\lambda/2$	$Z_{\text{in}} = Z_L, \quad n = 0, 1, 2, \dots$
Z_{in} of line of length $l = \lambda/4 + n\lambda/2$	$Z_{\text{in}} = Z_0^2 / Z_L, \quad n = 0, 1, 2, \dots$
Z_{in} of matched line	$Z_{\text{in}} = Z_0$
$ V_0^+ $ = amplitude of incident wave; $\Gamma = \Gamma e^{j\theta_t}$ with $-\pi < \theta_t < \pi$; θ_t in radians; $\Gamma_l = \Gamma e^{-j2\beta l}$.	

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Reflections on a Transmission Line

Reflections on a transmission line are caused by **impedance discontinuities**

Consider:

$$\Gamma = \frac{V_o^-}{V_o^+} = \frac{Z_L - Z_o}{Z_L + Z_o}$$

The reflection is nonzero whenever $Z_L \neq Z_o$. In general, a greater difference between the line and load impedances (a greater **mismatch**), will result in a greater percentage of the signal being reflected back toward the source.

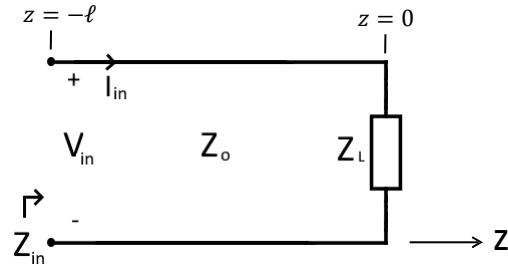
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Input Impedance

Input impedance is the impedance (ratio of total voltage to total current) seen at the input to a system.

Consider the impedance seen at the input port of this system:



where the variables are defined by:

$$V_{in} = V(-\ell) = \text{total voltage seen at the port}$$

$$I_{in} = I(-\ell) = \text{total current seen at the port}$$

$$Z_{in} = Z(-\ell) = \text{input impedance of the port} = V_{in}/I_{in}$$

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Input Impedance

$$\Gamma = \frac{V_o^-}{V_o^+} = \frac{Z_L - Z_o}{Z_L + Z_o}$$

Note that we also have the following relationships:

$$V_{in} = V(-\ell) = V_o^+ e^{\gamma \ell} + V_o^- e^{-\gamma \ell} = V_o^+ (e^{-\gamma \ell} + \Gamma e^{+\gamma \ell})$$

$$I_{in} = I(-\ell) = \frac{V_o^+}{Z_o} e^{+\gamma \ell} - \frac{V_o^-}{Z_o} e^{-\gamma \ell} = \frac{V_o^+}{Z_o} (e^{-\gamma \ell} - \Gamma e^{+\gamma \ell})$$

so that:

$$Z_{in} = \frac{Z_o (e^{+\gamma \ell} + \Gamma e^{-\gamma \ell})}{(e^{+\gamma \ell} - \Gamma e^{-\gamma \ell})} = \frac{Z_o \left(e^{+\gamma \ell} + \frac{Z_L - Z_o}{Z_L + Z_o} e^{-\gamma \ell} \right)}{\left(e^{+\gamma \ell} - \frac{Z_L - Z_o}{Z_L + Z_o} e^{-\gamma \ell} \right)}$$

which simplifies to:

$$Z_{in} = Z_o \frac{Z_L + jZ_o \tan(\beta \ell)}{Z_o + jZ_L \tan(\beta \ell)}$$

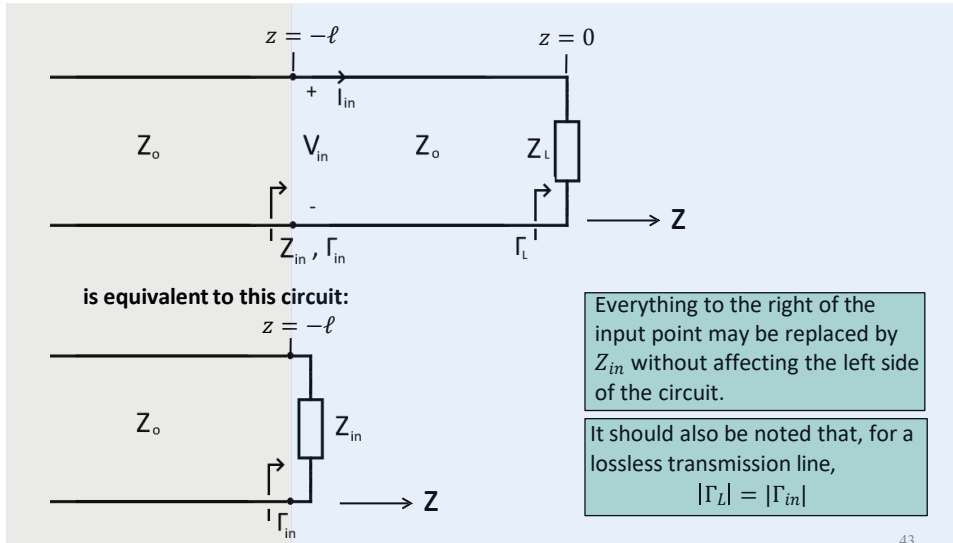
This is the input impedance – the impedance seen at the input to a line with characteristic impedance Z_o , if the line is terminated with load impedance Z_L .

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Input Impedance

The significance of the input impedance is that this circuit:



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Input Impedance – Special Cases

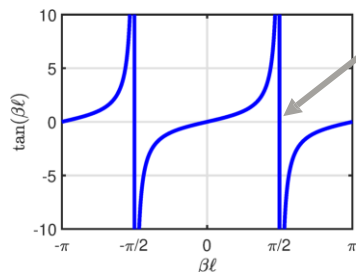
Case 1: short circuit termination ($Z_L = 0$)

Recall

$$Z_{in} = Z_o \frac{Z_L + jZ_o \tan(\beta\ell)}{Z_o + jZ_L \tan(\beta\ell)}$$

In this case,

$$Z_{in,sc} = jZ_o \tan(\beta\ell) = jZ_o \tan\left(\frac{2\pi\ell}{\lambda}\right)$$



Notes:

1) When $\ell = \frac{\lambda}{4}$, $\beta\ell = \frac{\pi}{2}$, and a short circuit ($Z_o = 0$) looks like an open circuit ($Z_{in} = \infty$)

2) The input impedance (Z_{in}) is inductive for $0 \leq \ell \leq \frac{\lambda}{4}$, and capacitive for $\frac{\lambda}{4} \leq \ell \leq \frac{\lambda}{2}$

3) The input impedance repeats every $\frac{\lambda}{2}$

$$\left(Z_{in}(z) = Z_{in}\left(z + \frac{\lambda}{2}\right) \right)$$

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Input Impedance – Special Cases

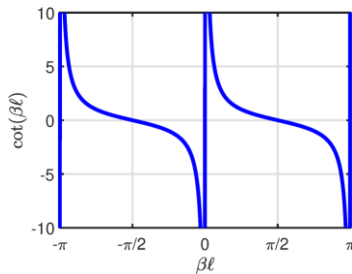
Case 2: open circuit termination ($Z_L = \infty$)

Recall

$$Z_{in} = Z_o \frac{Z_L + jZ_o \tan(\beta\ell)}{Z_o + jZ_L \tan(\beta\ell)}$$

In this case,

$$Z_{in,oc} = -jZ_o \cot(\beta\ell) = -jZ_o \cot\left(\frac{2\pi\ell}{\lambda}\right)$$



Notes:

- 1) When $\ell = \frac{\lambda}{4}$, $\beta\ell = \frac{\pi}{2}$, and an open circuit ($Z_o = \infty$) looks like a short circuit ($Z_{in} = 0$)
- 2) The input impedance (Z_{in}) is capacitive for $0 \leq \ell \leq \frac{\lambda}{4}$, and inductive for $\frac{\lambda}{4} \leq \ell \leq \frac{\lambda}{2}$
- 3) The input impedance repeats every $\frac{\lambda}{2}$

$$\left(Z_{in}(z) = Z_{in}\left(z + \frac{\lambda}{2}\right) \right)$$

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Input Impedance – Special Cases

Case 3: Matched load termination ($Z_L = Z_o$)

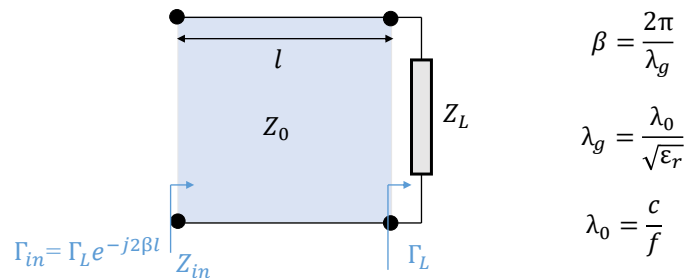
Recall

$$Z_{in} = Z_o \frac{Z_L + jZ_o \tan(\beta\ell)}{Z_o + jZ_L \tan(\beta\ell)}$$

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“Lossless” TL Summary

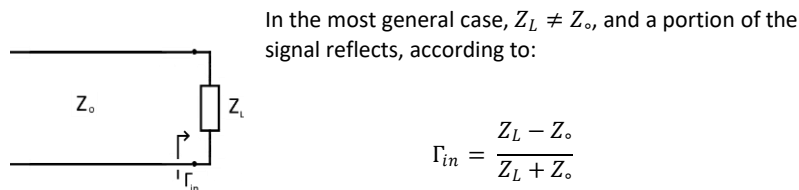


	Z_L	Z_{in}	Γ_L	VSWR
Load Z_L	Z_L	$= Z_0 \frac{Z_L + jZ_0 \tan(\beta l)}{Z_0 + jZ_L \tan(\beta l)}$	$= \frac{Z_L - Z_0}{Z_L + Z_0}$	$= \frac{1 + \Gamma }{1 - \Gamma }$
Matched (if $Z_0 = 50\Omega$)	50Ω	$= Z_0$	0	1
Open Circuit (OC)	∞	$= -jZ_0 \cot(\beta l)$	+1	∞
Short Circuit (SC)	0Ω	$= jZ_0 \tan(\beta l)$	-1	∞

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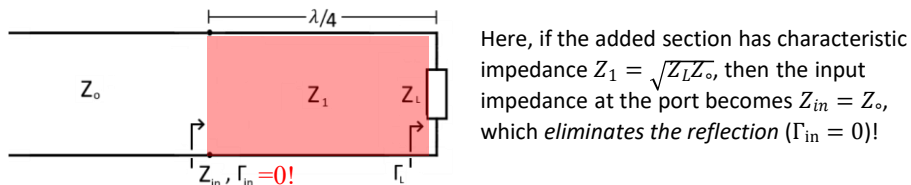
Quarter-Wave Transformer

Consider this circuit: a load impedance of Z_L is being driven using a line with impedance Z_0 .



To reduce the reflection, an engineer may choose to add a **matching network**.

One such network is a **quarter-wave transformer**, which adds an extra quarter-wave-long section of transmission line before the load, like this:

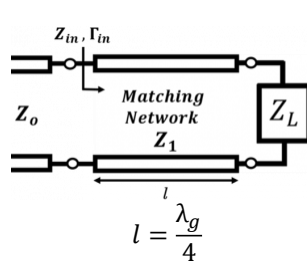


***Note: for this to work, Z_L must be purely real.

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Quarter-Wave Transformer



$$Z_{in} = \frac{Z_1^2}{Z_L}$$

For matching, $Z_{in} = Z_o$.

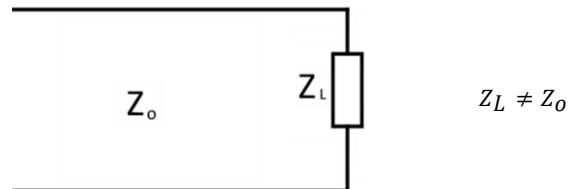
$$Z_1 = \sqrt{Z_o Z_L}$$

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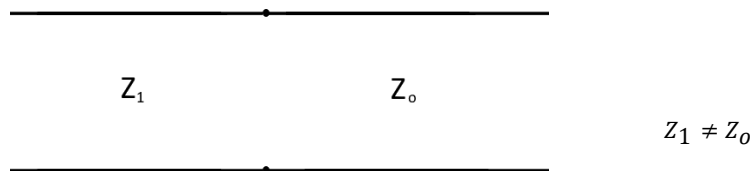
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Characteristic Impedance Mismatch

So far, we have discussed impedance discontinuities arising from a terminating load impedance that differs from the line impedance.



Reflections will also occur at line impedance discontinuities, as shown in the circuit below, where the line impedance value changes from Z_1 to Z_o .



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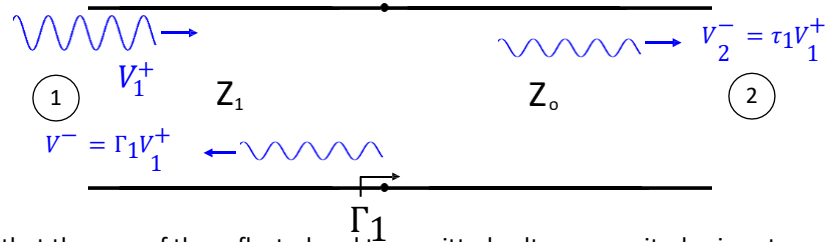
Characteristic Impedance Mismatch

If a wave is input from port 1, at the left (V_1^+), it will experience an instantaneous reflection at the discontinuity. The reflected wave (V_1^-) will have a magnitude of $\Gamma_1 V_1^+$, where:

$$\Gamma_1 = \frac{Z_o - Z_1}{Z_o + Z_1}$$

A portion of the wave will also be transmitted into the Z_o line, with voltage magnitude calculated using the **transmission coefficient** τ_1 , as $\tau_1 V_1^+$, where:

$$\tau_1 = 1 + \Gamma_1$$



***Notice that the sum of the reflected and transmitted voltage magnitudes is not equal to the input voltage magnitude. This is because it is **power** that is conserved, **not voltage**.

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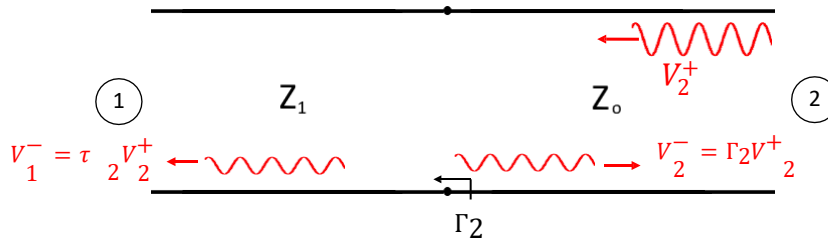
Characteristic Impedance Mismatch

Similarly, if a wave is input from the right (V_2^+), it will experience an instantaneous reflection at the discontinuity. The reflected wave (V_2^-) will have a magnitude of $\Gamma_2 V_2^+$, where:

$$\Gamma_2 = \frac{Z_1 - Z_o}{Z_1 + Z_o}$$

A portion of the wave will also be transmitted into the Z_1 line, with voltage magnitude calculated using the **transmission coefficient** τ_2 , as $\tau_2 V_2^+$, where:

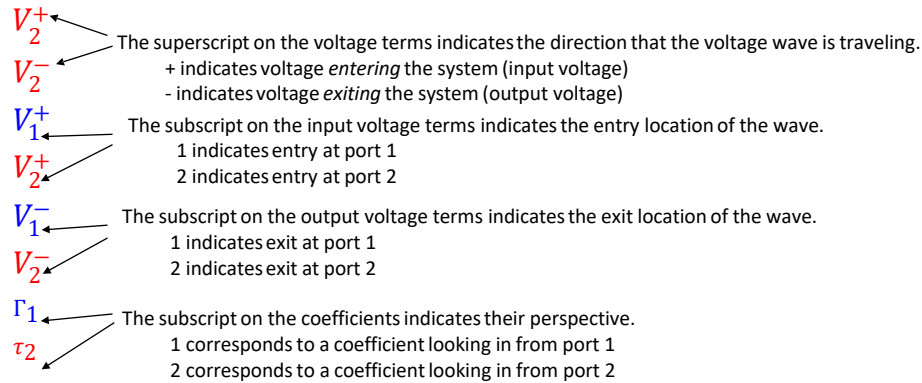
$$\tau_2 = 1 + \Gamma_2$$



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Notations



Note that **perspective of a coefficient matters!** Case in point:

$$\Gamma_1 = -\Gamma_2$$

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Sources

- The material presented herein is from the following sources:
 - Ulaby Chapter 2.1 to 2.8
 - “Elements of Electromagnetics,” by Matthew N.O Sadiku, 5th ed. (2010)
 - “Engineering Electromagnetics,” by Nathan Ida, 3rd ed. (2015)
 - “Microwave Engineering,” by David Pozar, 4th ed. (2012)
 - Ansys Innovation Courses and [Playlist](#)

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